Simulated Data Statement (Deliverable D)

Project: Predicting 30-day hospital readmission risk in diabetic patients, FUSE AI-201, Group 2

Statement

This project does not use simulated or synthetic data to train, tune, test or evaluate the model. All modelling and all reported results use one real, public dataset. A small number of hand-made input values were created only to test error handling; they are listed below and are not used as evidence of model performance.

Data source

Item	Detail
Dataset	*Diabetes 130-US Hospitals for Years 1999–2008*
Provider	UCI Machine Learning Repository, dataset 296
Link	https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008
Citation	Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). *Diabetes 130-US hospitals for years 1999–2008* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J
Size	101,766 hospital stays (71,518 patients), 50 columns; 99,340 stays used after exclusions
Licence	CC BY 4.0 (stated on the UCI page)
Privacy	De-identified research data. It contains no names and no contact details. We did not try to re-identify anyone and used no other data about these patients. The model never uses `patient_nbr` or `encounter_id` as inputs
Limits	US hospitals, 1999–2008; results from this data say nothing certain about today's hospitals or other countries

Where made-up or non-real values appear in the project

Where	What it is	Is it simulated data?
Edge-case tests (notebook Section 8.2, `tests/`, `docs/evidence/api_session.txt`)	Hand-made inputs such as a 60-day stay, 300 medications, 90 prior inpatient visits, race "Martian", insulin "Sideways", or text in a number field	**Yes, hand-made test inputs.** They are not patients. They exist only to check that invalid or unusual input produces warnings and no crash. Their scores are not performance results
Sample cases in the Streamlit app and API sample request	Three real, de-identified rows from our held-out **test set** (typical, higher-risk, lower-risk)	**No.** They are real records from the dataset, without identifiers
Default values in the app form	Illustrative starting values so the form is filled in	**Not patient data.** They are not a real person and not used for any result
Threshold and example values in	Proposed alert levels based on our	**No simulated data.** No

Where	What it is	Is it simulated data?
the monitoring plan	test results	production data exists, so none is simulated

What we do *not* claim

- We do not claim any real-world or production performance. All results come from a held-out part of the 1999–2008 dataset.
- We do not present the edge-case test outputs as evidence of accuracy.
- The prototype has not been validated on real patients and is not a medical tool.

Why no simulated data was needed

A real, public, de-identified dataset large enough to train and evaluate the model was available, so no data had to be invented. No live hospital data was available or appropriate for a student project.